CST519 - Secure Software Engineering

CST 519 (Secure Software Engineering) Syllabus

Course Information

Course Details

- **Course Title:** Secure Software Engineering
- **Course Number:** CST 519
- **Meeting Location:** Physical
- **Units:** 2

Course Description

This course explores principles and practices for designing, developing, and maintaining secure software systems. Topics may include secure software design, vulnerability analysis, threat modeling, secure coding, software supply chain security, and security testing.

Materials

- Course textbook: *Threat Modeling: A Practical Guide for Development Teams*, by Izar Tarandach and Matthew Coles (O'Reilly, 2020). Additional readings and tools will be provided through Canvas.

Technology Requirements

Students must have access to: - A computer (not "chromebook" or similar) with reliable internet connection - Access to Canvas learning management system

Course Objectives

At the end of this class, you should be able to:

1. Identify common software vulnerabilities and explain their likely causes and consequences.
2. Apply threat-modeling techniques to identify and prioritize security risks in a software system.
3. Design and implement secure software using established secure-coding and defensive-design practices.
4. Evaluate software, dependencies, and development workflows for security weaknesses and recommend mitigations.

Grading

There are three groups of assignments, briefly described in the table below.
Getting below a 40% in any of these three groups will result in failing the class.

Assignment Kind	Value
Projects	35%
Exams	35%
Participation	30%

Assignment Kind	Value

Projects

Projects will take the form of short, practical security-engineering exercises. These may include threat models, secure code reviews, vulnerability analysis, security-test plans, dependency or software-supply-chain assessments, and improvements to an existing small application.

Exams

Exams will cover material from class and from any assigned reading, as well as draw from knowledge gained during projects.

Participation

Class is expected to be highly interactive, with students actively answering and asking questions, as well as participating in discussion groups.
Therefore, attendance is required, as well as participation in all in-class activities.
These activities may take the form of in-class discussions, brainstorming sessions, stand-ups, as well as making active progress on projects during class time.

Grade Assignment

Grades as assigned based on the standard ranges, which are outlined below.

Grade Range	Letter Grade
$[97, \infty)$	A+
$[93, 97)$	A
$[90, 93)$	A-
$[87, 90)$	B+
$[83, 87)$	B
$[80, 83)$	B-
$[77, 80)$	C+
$[73, 77)$	C
$[70, 73)$	C-
$[60, 70)$	D
$[0, 60)$	F

Syllabus Change Policy

This syllabus is subject to change at the instructor's discretion to enhance learning or accommodate unforeseen circumstances. Students will be notified of any substantive changes (such as changes to due dates, point values of assignments, or major policy modifications) via Canvas announcement and/or email at least one week in advance when possible.

The process for syllabus changes:

1. Instructor identifies need for change
2. Revised syllabus section is prepared
3. Students are notified via Canvas and email
4. Change takes effect after notification period

Students are responsible for staying current with any announced changes to the syllabus.